

Docking Universal - Ligand-Free Cavity Report

Report generator version 0.6.4

Target: 5KRH

Interpretive status: exploratory site selection without a bound-ligand pose-recovery control

1. Configured docking protocol

This section records the settings used for this exploratory study. No target-specific bound-ligand pose-recovery control was available.

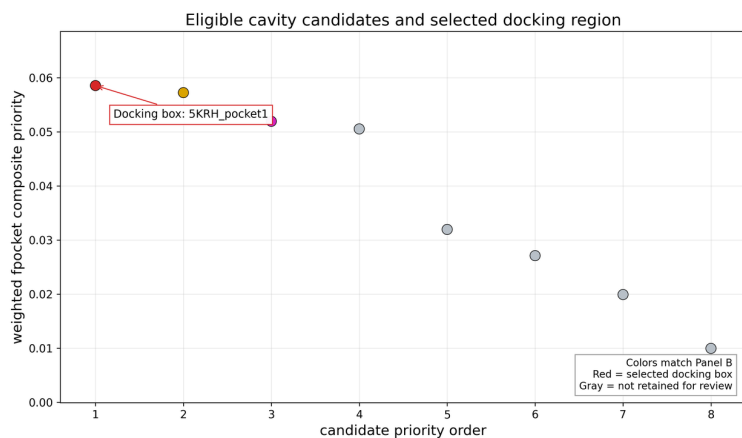
Parameter	Configured value
Validation status	Site-guided exploratory protocol; not evaluated by bound-ligand control
Engine	vina
Engine version	recorded when screening runs
Exhaustiveness	16
Modes per job	15
Energy range	8.0 kcal/mol
Independent seeds	5
Receptor	5KRH.pdbqt
Docking box	5KRH_pocket1.conf

2. Exploratory pocket configuration

No bound-ligand pose-recovery control was available. In its place, this section records how fpocket-generated cavity hypotheses were filtered and which docking box was selected. This documents site selection, but it does not validate the biological site or the accuracy of docked poses.

Pocket configuration	Recorded value
Interpretive status	Exploratory; not evaluated by a bound-ligand pose-recovery control
Selected docking pocket	pocket2_atm.pdb
fpocket eligibility threshold used	0.1
fpocket pocket score (Figure 1A)	0.1971
Composite selection priority	0.058558
fpocket druggability descriptor	0.506
Docking-box center (A)	-10.877000, -17.646000, -27.092000
Docking-box dimensions (A)	26 x 26 x 26

A



B

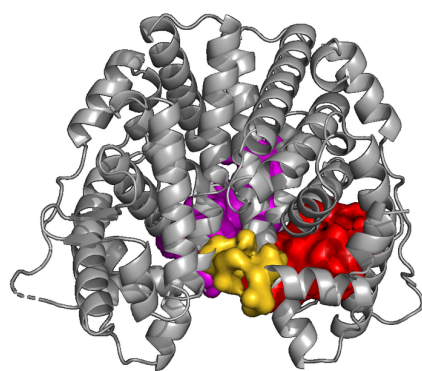


Figure 1. Exploratory pocket analysis. (A) Weighted fpocket composite priority for all candidates that passed the score and geometry eligibility filters, ordered by that priority. The weighting is $fpocket\ pocket\ score \times \exp(-distance\ from\ the\ protein\ centroid / 10\ \text{\AA})$; it favors interior pockets. The colored candidates are the three retained for review and color-match Panel B; gray candidates were not retained for review. The unweighted fpocket pocket score is recorded in the configuration table. (B) The retained pocket hypotheses are shown as color-matched surfaces on the receptor. These geometric pocket surfaces are not experimentally observed molecular surfaces, and fpocket scores are not binding-affinity estimates.

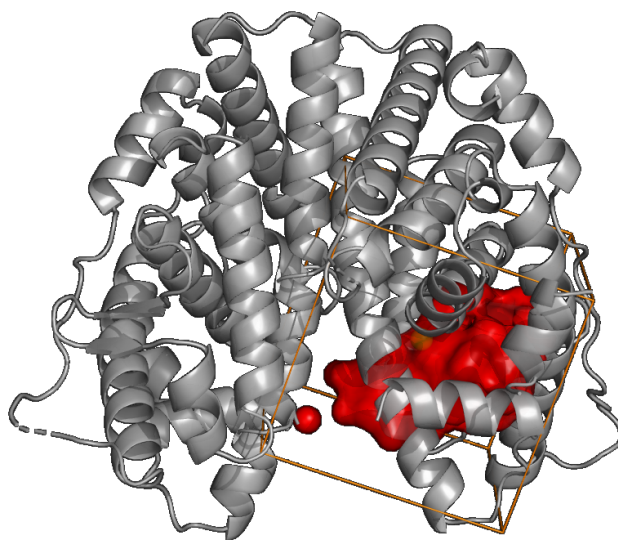


Figure 2. Chosen pocket and docking box. The receptor is shown as a gray cartoon, the selected alpha-sphere-derived pocket surface in translucent red, the selected center in yellow, and the docking box as an orange wireframe. Red matches the selected candidate in Figure 1. This records the region chosen to proceed; it does not establish that the pocket is biologically correct.

3. Reproducibility, software, and references

fpocket supplied ligand-free geometric cavity candidates and descriptors; the report records the selected docking box separately. Receptor preparation path: user-approved removal of unmatched receptor components after safe preparation fallbacks failed. No ligand docking, scoring, pose clustering, or interaction analysis was performed. PyMOL produced the structural panels, and ReportLab assembled this PDF.

PDBFixer audit: added 127 missing side-chain heavy atoms; detected but did not add 4 terminal atoms. The repaired intermediate was rejected by strict Meeko, so these changes were not used in the final receptor; the user explicitly approved removal of unmatched components from the filtered original.

User-approved receptor component removal: safe preparation fallbacks failed, and the user explicitly approved Meeko's removal of unmatched components. The receptor model was changed: 31 standard amino-acid residues were removed. **HIGH-SEVERITY STRUCTURAL MODIFICATION. High-severity structural warning:** standard protein/peptide residues, not merely solvent or optional hetero components, were omitted from the final receptor. Inspect the complete retained removal manifest and log; a target-matched bound-ligand control is required before prospective screening.

Software versions used for this report

Result element	Software	Version used
Scientific workflow	Docking Universal	0.6.3
Ligand-free cavity detection	fpocket	4.2.2
Receptor preparation	Meeko	0.7.1
Conditional conservative receptor repair	PDBFixer	1.11.0
3D rendering	PyMOL	3.0.0
Plots	Matplotlib	3.9.4
PDF generation	ReportLab	5.0.0
Scientific workflow runtime	Python	3.9.23

Scientific and software references

1. Le Guilloux V, Schmidtke P, Tuffery P. Fpocket: an open source platform for ligand pocket detection. BMC Bioinformatics. 2009;10:168. <https://doi.org/10.1186/1471-2105-10-168>
2. Santos-Martins D, He Y, Eberhardt J, et al. Meeko: molecule parameterization and software interoperability for docking and beyond. J Chem Inf Model. 2025;65:13045-13050. <https://doi.org/10.1021/acs.jcim.5c02271>
3. PDBFixer: a tool for preparing PDB files for molecular simulation (version recorded above). <https://github.com/openmm/pdbfixer>
4. Eastman P, Swails J, Chodera JD, et al. OpenMM 7: Rapid development of high performance algorithms for molecular dynamics. PLoS Comput Biol. 2017;13:e1005659. <https://doi.org/10.1371/journal.pcbi.1005659>
5. The PyMOL Molecular Graphics System, Version 3.0.0, Schrodinger, LLC. <https://www.pymol.org/support.html>